

PERRY, IA, USA

WMO: 720412

Lat: 41.828N

Lon: 94.160W

Elev: 1013

StdP: 14.17

Time Zone: -6.00 (NAC)

Period: 08-19

WBAN: 54943

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.6	(c) -1.5	(d) -14.6	(e) 2.6	(f) -6.6	(g) -8.7	(h) 3.6	(i) -0.5	(j) 30.5	(k) 24.9	(l) 27.7	(m) 21.8	(n) 7.8	(o) 310	(p) 0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 20.6	(c) 92.8	(d) 72.8	(e) 90.1	(f) 73.5	(g) 87.2	(h) 72.5	(i) 79.0	(j) 86.5	(k) 77.1	(l) 84.9	(m) 75.2	(n) 82.8	(o) 12.3	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.0	146.1	84.1	74.8	135.5	81.2	73.0	127.1	79.3	43.3	86.5	41.2	84.5	39.4	82.6	85.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
26.4	23.6	20.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
			DB												
			WB	-16.5	82.3	6.0	2.3	-20.8	84.0	-24.3	85.3	-27.7	86.6	-32.0	88.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.6	21.3	22.9	37.7	50.4	62.5	72.0	73.4	71.2	65.6	52.3	38.3	26.5
	DBStd	21.15	12.62	12.14	11.61	9.53	9.20	6.35	6.07	5.20	8.24	9.12	10.42	11.14
	HDD50	3369	890	761	410	109	12	1	0	0	4	83	369	730
	HDD65	6502	1355	1180	847	445	157	14	6	8	88	406	801	1195
	CDD50	3240	0	1	30	121	400	660	726	657	473	153	19	0
	CDD65	897	0	0	2	8	80	223	267	200	106	11	0	0
	CDH74	8746	0	0	20	131	825	2159	2665	1723	1075	144	4	0
	CDH80	3157	0	0	2	33	301	826	1047	543	372	33	0	0
Wind	WSAvg	9.1	10.5	10.1	10.8	11.7	10.6	8.8	5.9	5.4	6.6	8.4	10.0	10.0
Precipitation	PrecAvg	35.9	1.1	1.1	2.2	3.7	4.9	5.4	4.8	4.3	3.1	2.4	1.9	1.2
	PrecMax	46.3	2.0	1.9	5.1	8.0	9.1	11.3	11.3	8.9	6.5	7.0	4.6	2.5
	PrecMin	23.7	0.4	0.1	0.1	0.7	1.4	0.7	1.3	0.8	1.0	0.4	0.1	0.0
	PrecStd	6.2	0.4	0.5	1.3	1.9	2.0	2.8	2.6	2.3	1.6	1.8	1.3	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.8	61.0	77.1	85.1	94.0	95.1	98.6	93.8	92.7	84.4	72.9	56.4
		MCWB	44.1	53.0	64.0	65.3	65.9	74.6	75.6	73.0	70.4	63.7	56.2	47.4
	2%	DB	45.7	49.8	69.3	77.3	87.9	91.2	92.8	89.6	88.4	77.7	64.2	49.9
		MCWB	41.0	44.9	57.0	59.1	67.3	72.7	75.9	73.8	72.7	60.6	53.0	45.2
	5%	DB	40.8	43.5	62.6	72.2	82.2	88.3	89.8	85.7	84.2	72.8	59.0	45.2
		MCWB	37.6	40.0	55.3	56.3	64.4	71.7	76.8	74.0	69.7	59.7	49.7	40.9
	10%	DB	37.0	37.7	55.3	67.8	78.5	85.1	86.0	82.2	80.7	68.3	54.3	40.3
		MCWB	34.8	35.7	48.5	54.6	63.0	70.0	74.0	71.6	67.2	57.4	47.4	37.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.4	52.6	66.7	67.2	73.4	80.2	82.9	80.3	77.2	70.0	59.3	51.5
		MCDB	50.8	59.5	75.6	75.4	84.2	89.8	89.7	87.2	87.0	73.8	66.6	52.3
	2%	WB	41.2	45.6	60.5	62.7	70.4	77.2	80.1	77.4	75.2	66.2	55.6	45.4
		MCDB	45.5	49.1	66.0	71.7	79.8	86.0	87.2	83.6	83.5	73.2	64.3	49.0
	5%	WB	38.0	40.0	54.8	59.5	68.4	75.0	77.7	75.3	73.2	62.8	51.3	41.2
		MCDB	40.7	43.5	61.1	69.8	77.3	83.4	85.6	80.6	79.7	68.2	56.9	44.3
	10%	WB	35.1	36.0	49.5	56.4	66.3	73.0	75.9	73.5	71.0	59.3	47.4	37.6
		MCDB	36.9	38.0	55.6	64.7	75.3	81.4	84.0	80.3	77.1	66.2	54.1	40.3
Mean Daily Temperature Range	5% DB	MDBR	17.6	17.1	17.8	21.9	20.8	20.1	20.6	20.5	23.9	22.4	19.3	16.0
		MCDBR	21.9	21.1	28.9	30.9	26.6	23.5	23.4	23.5	25.7	28.6	27.5	21.9
		MCWBR	17.1	16.1	17.8	16.0	10.3	10.3	11.0	11.5	10.9	14.2	16.1	16.3
	5% WB	MCDBR	20.4	21.0	25.7	24.7	21.0	20.2	20.8	19.7	20.7	22.2	24.5	19.3
		MCWBR	16.0	16.3	16.7	14.8	10.5	11.4	11.4	11.8	11.0	14.5	16.4	15.1
Clear-Sky Solar Irradiance	taub	0.293	0.308	0.327	0.382	0.411	0.421	0.435	0.422	0.388	0.342	0.298	0.289	
	taud	2.405	2.370	2.410	2.276	2.266	2.284	2.227	2.274	2.321	2.442	2.504	2.437	
	Ebn at Noon	272	285	292	281	275	271	266	266	267	269	269	264	
	Edh at Noon	25	30	33	41	42	42	43	40	36	28	23	23	
All-Sky Solar Radiation	RadAvg	612	933	1205	1512	1787	1984	2015	1736	1444	977	665	506	
	RadStd	57	64	116	121	137	176	140	116	103	117	52	44	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page